

# RTH Gap Trading Playbook

## Consolidated Statistical Edge Report & Morning Decision System

**Generated:** January 23, 2026

**Data Range:** 2006-2026 (4,000-5,000 sessions per ticker)

**Tickers Analyzed:** NQ, ES, YM, RTY, CL, GC

# Part 1: Morning Decision Flowchart

09:25 ET - PRE-MARKET CHECKLIST

STEP 1: MEASURE THE GAP

Gap % = (RTH Open - Prior RTH Close) / Prior RTH Close × 100

GAP < 0.15%  
(NOISE)  
Fill Rate: 85%+  
Action: FADE

GAP 0.15-0.45%  
(CONFLICT ZONE)  
Fill Rate: 55%  
Action: FILTER

GAP > 0.45%  
(SIGNAL)  
Fill Rate: 35%  
Action: DEFEND

STEP 2: CHECK GLOBEX CONTEXT

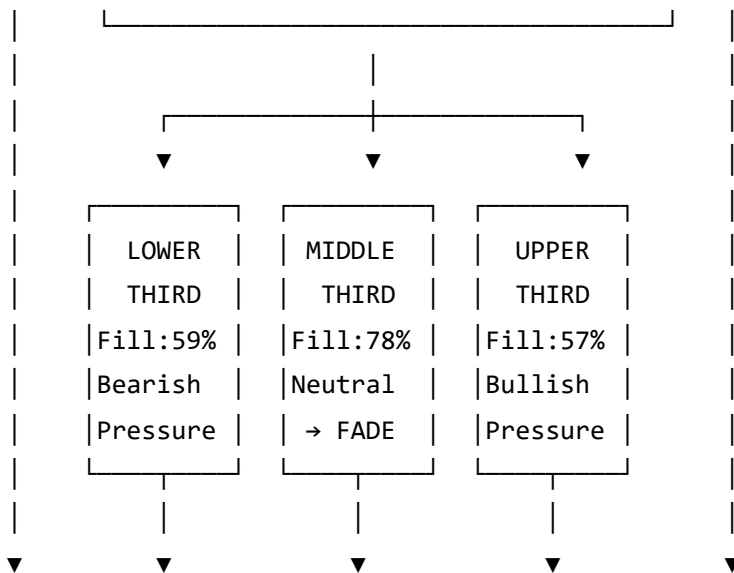
NARROW  
<50%ATR  
Fill:76%  
→ FADE

NORMAL  
50-100%  
ATR  
Fill:61%  
→FILTER

WIDE  
>100%  
ATR  
Fill:41%  
→DEFEND

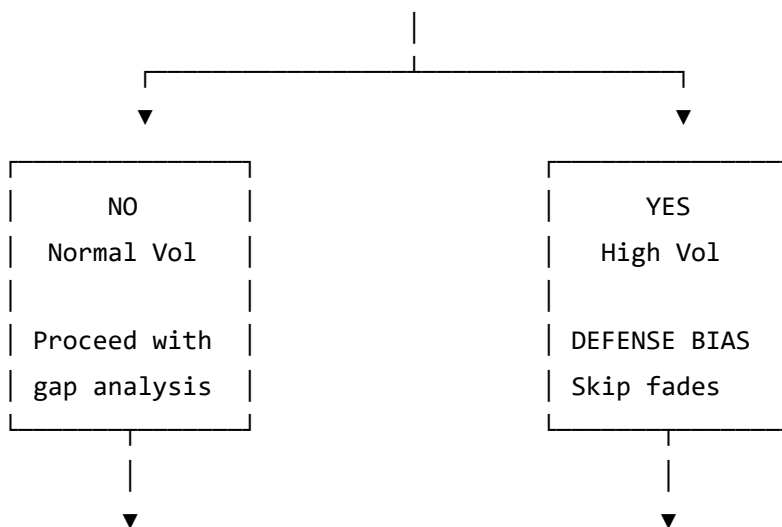
STEP 3: RTH OPEN POSITION

Where did RTH open within Globex?

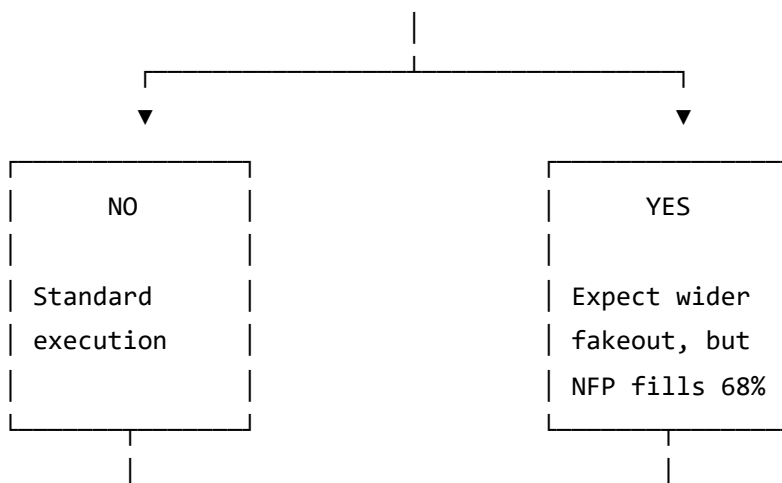


#### STEP 4: VOLATILITY REGIME CHECK

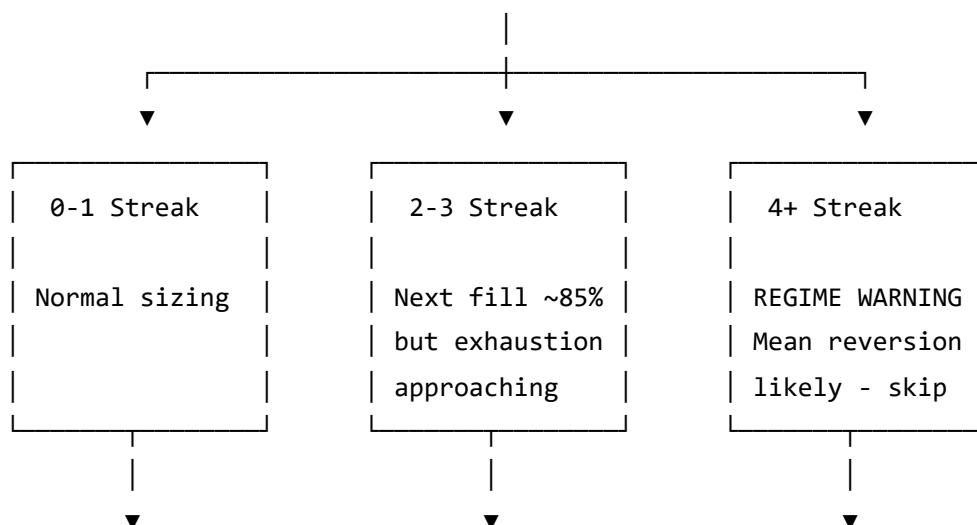
Is VVIX > 110? OR Is ATR in "High" bucket?



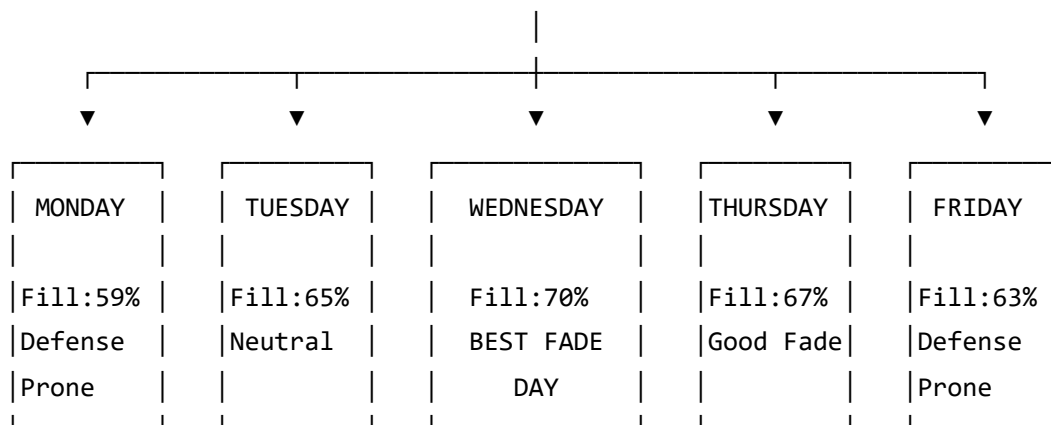
#### STEP 5: NEWS CHECK - Is there 8:30 AM News? (NFP/CPI/GDP/Retail)



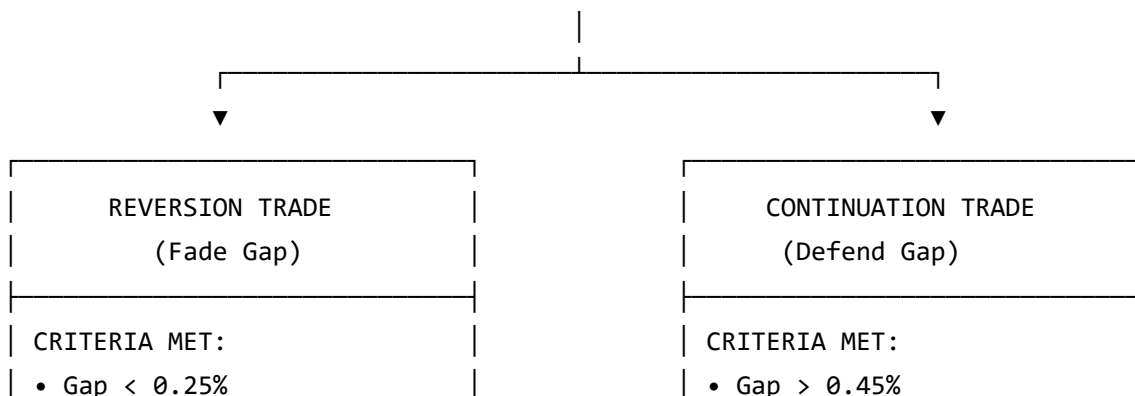
STEP 6: STREAK CHECK - How many consecutive fills/defends?



STEP 7: DAY OF WEEK FILTER



FINAL DECISION GATE



|                                                                                                                                                                                                 |                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Narrow Globex</li> <li>• Middle Third Open</li> <li>• Normal/Low Vol</li> <li>• Tue/Wed/Thu</li> <li>• No 4+ fill streak</li> </ul>                    | <ul style="list-style-type: none"> <li>• Wide Globex</li> <li>• Upper/Lower Third Open</li> <li>• High Vol/ATR</li> <li>• Monday or Friday</li> <li>• OBR Open Type</li> </ul>              |
| <p>EXECUTION:</p> <ul style="list-style-type: none"> <li>• Entry: Fade at open</li> <li>• Stop: 1.0x gap beyond open</li> <li>• Target: Prior RTH close</li> <li>• Time Stop: 30 min</li> </ul> | <p>EXECUTION:</p> <ul style="list-style-type: none"> <li>• Entry: With gap direction</li> <li>• Stop: Prior day close</li> <li>• Target: 1.5-2x gap size</li> <li>• No time stop</li> </ul> |
| <p>EXPECTED WIN RATE: 88-91%<br/>(Perfect Reversion Setup)</p>                                                                                                                                  | <p>EXPECTED WIN RATE: 60-70%<br/>(Defense Setup)</p>                                                                                                                                        |



|                      |
|----------------------|
| 09:30 ET - EXECUTION |
|----------------------|



|                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>STEP 8: THE 15-MINUTE MOAT CHECK (Confirmation)</p> <p>After first 15-min candle:</p> <ul style="list-style-type: none"> <li>• If prior extreme HOLDS → Defense confirmed, exit fade or join trend</li> <li>• If prior extreme BREAKS → Reversion confirmed, hold for fill</li> </ul> |
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|                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>STEP 9: PARTIAL FILL MANAGEMENT</p> <ul style="list-style-type: none"> <li>• At 50% fill: 82% chance of full fill - HOLD</li> <li>• At 75% fill: 90% chance of full fill - HOLD</li> <li>• If stalled at 50% for 15+ min: Consider reducing size</li> <li>• If not 50% filled by 30 min: Reassess or exit</li> </ul> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Part 2: Cross-Ticker Statistical Summary

## Overall Fill Rates & Best Setups

| Ticker | Base Fill | Perfect Reversion | Lift   | Sample | Trend Continuation | Defense Rate |
|--------|-----------|-------------------|--------|--------|--------------------|--------------|
| ES     | 64.5%     | 91.1%             | +26.7% | 946    | 31.6%              | 68.4%        |
| YM     | 63.3%     | 91.1%             | +27.8% | 854    | 30.2%              | 69.8%        |
| NQ     | 66.8%     | 88.9%             | +22.1% | 1085   | 36.9%              | 63.1%        |
| RTY    | 65.4%     | 90.6%             | +25.2% | 297    | 38.5%              | 61.5%        |
| CL     | 55.4%     | 88.3%             | +32.9% | 205    | 39.4%              | 60.6%        |
| GC     | 51.5%     | 78.9%             | +27.4% | 722    | 22.3%              | 77.7%        |

**Recommendation:** ES or YM for highest "Perfect Reversion" rate (91.1%). NQ for fastest fills.

## Fill Probability by Gap Size

| Bucket              | NQ    | ES    | YM    | RTY   | CL    | GC    |
|---------------------|-------|-------|-------|-------|-------|-------|
| Very Small (<0.07%) | 93.5% | 94.3% | 94.2% | 94.7% | 88.2% | 79.2% |
| Small (0.07-0.15%)  | 79.0% | 82.7% | 83.2% | 88.8% | 86.5% | 76.3% |
| Medium (0.15-0.25%) | 68.4% | 67.7% | 67.8% | 82.5% | 77.7% | 62.0% |
| Large (0.25-0.45%)  | 54.1% | 56.0% | 57.1% | 65.7% | 74.1% | 47.9% |
| Very Large (>0.45%) | 38.6% | 35.0% | 33.9% | 44.6% | 47.1% | 21.9% |

**Key Insight:** Gaps <0.15% fill 85%+ across all equity indices. Gaps >0.45% defend 60%+ of the time.

# Time-to-Fill Analysis

## Speed of Fill by Ticker

| Ticker | % Filled <15m | % Filled <30m | Median Fill (Small Gap) |
|--------|---------------|---------------|-------------------------|
| RTY    | 32.5%         | 43.4%         | 2 min                   |
| NQ     | 33.7%         | 45.2%         | 11 min                  |
| YM     | 29.4%         | 38.4%         | 9 min                   |
| ES     | 28.3%         | 39.0%         | 16 min                  |
| CL     | 16.4%         | 23.3%         | 0 min                   |
| GC     | 16.7%         | 25.6%         | 3 min                   |

**Key Insight:** If not 50% filled in 30 minutes, reassess the trade. High-conviction fills happen fast.

## Fill Time by Gap Size (NQ Example)

| Gap Size   | Median Fill Time | % Filled <15m |
|------------|------------------|---------------|
| Very Small | 1 min            | 83.5%         |
| Small      | 11 min           | 54.8%         |
| Medium     | 25 min           | 36.2%         |
| Large      | 50 min           | 20.0%         |
| Very Large | 100 min          | 7.5%          |

## Partial Fill Conditional Probabilities

**Critical for Trade Management:**

| If Price Reaches... | NQ    | ES    | YM    | RTY   |
|---------------------|-------|-------|-------|-------|
| 25% of gap          | 75.2% | 73.0% | 72.6% | 73.8% |

| If Price Reaches... | NQ    | ES    | YM    | RTY   |
|---------------------|-------|-------|-------|-------|
| 50% of gap          | 82.8% | 81.4% | 81.3% | 82.0% |
| 75% of gap          | 90.8% | 89.7% | 90.1% | 90.4% |

**Key Insight:** Once price reaches 50% of the gap, hold for full fill (82% probability). At 75%, it's 90%+.

## Consecutive Day Streak Analysis

### Fill Probability After N Consecutive Fills

| Streak  | NQ    | ES    | YM    | RTY   |
|---------|-------|-------|-------|-------|
| 1 Fill  | 68.6% | 66.6% | 65.0% | 66.2% |
| 2 Fills | 81.9% | 79.7% | 76.4% | 77.5% |
| 3 Fills | 90.1% | 87.9% | 85.9% | 84.0% |
| 4 Fills | 96.9% | 94.7% | 94.9% | 92.5% |
| 5 Fills | 98.6% | 94.8% | 96.7% | 98.0% |

**Key Insight:** After 3+ consecutive fills, a defense day is statistically due. Consider sizing down or skipping.

### Prior Day Impact

| Prior Day Outcome | NQ Fill Today | ES Fill Today | YM Fill Today |
|-------------------|---------------|---------------|---------------|
| Prior Defended    | 69.4%         | 68.1%         | 66.5%         |
| Prior Filled      | 65.5%         | 62.4%         | 61.4%         |

**Key Insight:** Defense days tend to cluster. If yesterday defended, today has slightly higher fill probability.



# Globex Range Context

## Fill Rate by Overnight Range Size

| Globex Range      | NQ    | ES    | YM    | RTY   |
|-------------------|-------|-------|-------|-------|
| Narrow (<50% ATR) | 75.8% | 76.3% | 77.4% | 75.7% |
| Normal (50-100%)  | 62.8% | 61.2% | 59.2% | 61.3% |
| Wide (>100% ATR)  | 42.6% | 41.2% | 38.9% | 42.1% |

**Key Insight:** Narrow overnight sessions = highest reversion probability (~76-77%).

## Fill Rate by RTH Open Position in Globex Range

| Position     | NQ    | ES    | YM    | RTY   |
|--------------|-------|-------|-------|-------|
| Lower Third  | 63.3% | 59.2% | 58.2% | 61.3% |
| Middle Third | 78.0% | 78.8% | 78.7% | 76.1% |
| Upper Third  | 60.6% | 56.6% | 55.7% | 60.1% |

**Key Insight:** Middle third opens have highest fill rate (~78%). Extreme positions signal directional bias.

# Day of Week Analysis

| Day       | NQ Fill | ES Fill | YM Fill | Best Strategy |
|-----------|---------|---------|---------|---------------|
| Monday    | 60.6%   | 58.4%   | 57.9%   | Defense Prone |
| Tuesday   | 68.0%   | 65.0%   | 62.3%   | Neutral       |
| Wednesday | 70.0%   | 69.2%   | 69.7%   | Best Fade Day |
| Thursday  | 69.0%   | 66.4%   | 66.9%   | Good Fade Day |
| Friday    | 66.0%   | 62.7%   | 59.1%   | Defense Prone |

**Key Insight:** Mid-week (Tue-Thu) for fades. Monday/Friday gaps that hold tend to continue.

## MAE/MFE Precision (Stop Placement)

### Fakeout Before Fill (MFE in Gap Direction)

| Ticker | Median Fakeout | Stop Recommendation |
|--------|----------------|---------------------|
| RTY    | 87.6%          | 1.0x gap size       |
| YM     | 85.0%          | 1.0x gap size       |
| NQ     | 83.3%          | 1.0x gap size       |
| ES     | 77.8%          | 0.9x gap size       |
| CL     | 66.7%          | 0.8x gap size       |
| GC     | 57.6%          | 0.7x gap size       |

**Key Insight:** ES has the tightest fakeout profile - best for defined risk. Place stops beyond 80-100% of gap size.

### Trend Extension (When Gap Defends)

| Ticker | Median Extension | Target Recommendation |
|--------|------------------|-----------------------|
| NQ     | 160 pts          | 1.5-2x gap size       |
| RTY    | 155 pts          | 1.5-2x gap size       |
| YM     | 147 pts          | 1.5-2x gap size       |
| ES     | 139 pts          | 1.5-2x gap size       |
| CL     | 115 pts          | 1.5x gap size         |
| GC     | 101 pts          | 1x gap size           |

# Volatility Regime Impact

## Fill Rate by ATR Bucket

| ATR Regime | NQ    | ES    | YM    |
|------------|-------|-------|-------|
| Low ATR    | 70.6% | 67.7% | 65.3% |
| Normal ATR | 66.8% | 63.6% | 63.2% |
| High ATR   | 63.1% | 62.1% | 61.5% |

**Key Insight:** High ATR = larger gaps, lower fill rates. Treat as breakaway signal.

## VIX Impact

| VIX Regime     | ES Fill | YM Fill |
|----------------|---------|---------|
| Low VIX        | 64.0%   | 71.7%   |
| Normal VIX     | 66.3%   | 67.0%   |
| High VIX (>25) | 65.3%   | 59.9%   |

**Key Insight:** High VIX widens the "moat" - gaps fill less frequently on YM especially.

# News Day Profiles

## 8:30 AM News Impact

| Event   | NQ Fill      | ES Fill      | Avg Gap Size |
|---------|--------------|--------------|--------------|
| No News | 66.7%        | 64.1%        | 0.32%        |
| NFP     | <b>71.3%</b> | <b>68.3%</b> | 0.30%        |
| CPI     | 65.0%        | 63.5%        | 0.35%        |
| GDP     | 68.8%        | 79.2%        | 0.23%        |

**Key Insight:** NFP has HIGH fakeouts but also HIGH fill rates. Don't fade early - wait for the flush.

## Part 3: The Two Optimal Setups

### Setup A: Perfect Reversion (91% Win Rate)

#### Entry Criteria (ALL must be true)

| Filter            | Requirement                  | Why                              |
|-------------------|------------------------------|----------------------------------|
| Gap Size          | 0.07% - 0.25%                | Small gaps are noise             |
| Globex Range      | Narrow (<50% ATR)            | Quiet overnight = mean reversion |
| RTH Open Position | Middle third of Globex       | No directional pressure          |
| Day of Week       | Tuesday, Wednesday, Thursday | Mid-week fills best              |
| ATR Regime        | Low or Normal                | High ATR = breakaway             |
| Streak            | No 4+ consecutive fills      | Regime exhaustion                |

#### Execution Parameters

| Parameter | Value                                 | Rationale                   |
|-----------|---------------------------------------|-----------------------------|
| Entry     | Fade at RTH open or first 5m pullback | Immediate execution         |
| Stop      | 1.0x gap size beyond open             | Survives 83% median fakeout |
| Target    | Prior RTH close (100% fill)           | Full gap fill               |
| Time Stop | 30 minutes                            | If not 50% filled, reassess |

#### Expected Performance

| Ticker | Win Rate | Sample Size | Risk:Reward       |
|--------|----------|-------------|-------------------|
| ES     | 91.1%    | 946 days    | ~9:1 EV at 1:1 RR |

| Ticker | Win Rate | Sample Size | Risk:Reward       |
|--------|----------|-------------|-------------------|
| YM     | 91.1%    | 854 days    | ~9:1 EV at 1:1 RR |
| NQ     | 88.9%    | 1085 days   | ~8:1 EV at 1:1 RR |
| RTY    | 90.6%    | 297 days    | ~9:1 EV at 1:1 RR |

## Setup B: Trend Continuation / Defense (68% Win Rate)

### Entry Criteria (ALL must be true)

| Filter       | Requirement             | Why                                |
|--------------|-------------------------|------------------------------------|
| Gap Size     | >0.45%                  | Large gaps are signal              |
| Open Type    | OBR (Outside Bar Range) | Already gapped through prior range |
| Globex Range | Wide (>100% ATR)        | Overnight momentum                 |
| ATR Regime   | High                    | Volatility expansion               |
| Day of Week  | Monday or Friday        | Trend days                         |

### Execution Parameters

| Parameter | Value                                   | Rationale            |
|-----------|-----------------------------------------|----------------------|
| Entry     | With gap direction after 15m moat holds | Confirmation         |
| Stop      | Prior RTH close                         | If fill starts, exit |
| Target    | 1.5-2x gap size extension               | Trending gaps run    |
| Time Stop | None                                    | Let it ride          |

### Expected Performance

| Ticker | Defense Rate | Sample Size | Best For             |
|--------|--------------|-------------|----------------------|
| GC     | 77.7%        | 878 days    | Most persistent gaps |

| Ticker | Defense Rate | Sample Size | Best For           |
|--------|--------------|-------------|--------------------|
| YM     | 69.8%        | 656 days    | Blue chip momentum |
| ES     | 68.4%        | 651 days    | Balanced           |

# Part 4: Quick Reference Cards

## Morning Checklist (Print This)

| 09:25 ET MORNING CHECKLIST                                                                         |
|----------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Gap Size: _____% (< 0.15 = Fade   > 0.45 = Defend)                        |
| <input type="checkbox"/> Globex Range: _____ ATR%<br>(< 50% = Fade   > 100% = Defend)              |
| <input type="checkbox"/> RTH Open Position: Lower / Middle / Upper Third<br>(Middle = Fade bias)   |
| <input type="checkbox"/> Day of Week: _____<br>(Tue-Thu = Fade   Mon/Fri = Defend)                 |
| <input type="checkbox"/> VVIX Level: _____ (> 110 = Defend bias)                                   |
| <input type="checkbox"/> ATR Regime: Low / Normal / High<br>(High = Defend bias)                   |
| <input type="checkbox"/> 8:30 News? Y / N (NFP = wait for flush then fade)                         |
| <input type="checkbox"/> Consecutive Fills: _____ days<br>(4+ = skip or size down)                 |
| DECISION: <input type="checkbox"/> FADE (Reversion) <input type="checkbox"/> DEFEND (Continuation) |
| TICKER: _____ SIZE: _____ STOP: _____ TARGET: _____                                                |

# Probability Lookup Table

| FILL RATE QUICK LOOKUP |       |       |       |       |
|------------------------|-------|-------|-------|-------|
| CONDITION              | NQ    | ES    | YM    | RTY   |
| Base Rate              | 66.8% | 64.5% | 63.3% | 65.4% |
| Perfect Setup          | 88.9% | 91.1% | 91.1% | 90.6% |
| Gap < 0.15%            | 86%   | 88%   | 88%   | 92%   |
| Gap > 0.45%            | 39%   | 35%   | 34%   | 45%   |
| Narrow Globex          | 76%   | 76%   | 77%   | 76%   |
| Wide Globex            | 43%   | 41%   | 39%   | 42%   |
| Middle Third           | 78%   | 79%   | 79%   | 76%   |
| Wednesday              | 70%   | 69%   | 70%   | 67%   |
| Monday                 | 61%   | 58%   | 58%   | 64%   |
| After 3 Fills          | 90%   | 88%   | 86%   | 84%   |
| At 50% Retrace         | 83%   | 81%   | 81%   | 82%   |
| At 75% Retrace         | 91%   | 90%   | 90%   | 90%   |

# Stop Placement Guide

| STOP PLACEMENT GUIDE                                  |                   |                 |
|-------------------------------------------------------|-------------------|-----------------|
| TICKER                                                | STOP (x Gap Size) | SURVIVES        |
| ES                                                    | 0.9x              | 78% of fakeouts |
| NQ                                                    | 1.0x              | 83% of fakeouts |
| YM                                                    | 1.0x              | 85% of fakeouts |
| RTY                                                   | 1.0x              | 88% of fakeouts |
| RULE: Place stop beyond median fakeout to avoid noise |                   |                 |
| ES wins here - tightest fakeouts = best defined risk  |                   |                 |



# Part 5: Ticker Selection Guide

## When to Trade Each Ticker

| Ticker | Best For                  | Avoid When       | Edge                              |
|--------|---------------------------|------------------|-----------------------------------|
| ES     | Defined risk fades        | High VIX spikes  | Tightest fakeout (78%), most data |
| YM     | Slow, managed fades       | Fast markets     | Slowest fills, best for scaling   |
| NQ     | Quick resolution          | Need tight stops | Fastest fills, highest base rate  |
| RTY    | Small gaps, fastest fills | Large gaps       | 92% fill on <0.15% gaps           |
| CL     | Avoid for fades           | Most conditions  | Too volatile, lowest edge         |
| GC     | Continuation plays        | Fading           | 78% defense on large gaps         |

## Final Recommendation

### For Defined Risk Reversion Trading:

- 1. **Primary:** ES - 91.1% win rate on perfect setup, tightest fakeout profile
- 2. **Secondary:** YM - Same win rate, slower fills give more management time
- 3. **Tertiary:** NQ - Slightly lower win rate but fastest resolution

### For Continuation/Defense Trading:

- 1. **Primary:** GC - 77.7% defense rate, most persistent gaps
- 2. **Secondary:** YM - Clean trend days on Monday/Friday

*Document generated from 20 years of futures data (2006-2026). Statistics are historical and do not guarantee future performance. Always use proper risk management.*